Expected Visiting Times of Interval Markov Chains

Jesper Nilsson, 457057

Abstract—test

1 Introduction

Thema einführen

2 Preliminaries

2.1 Notation

We denote the entry in the i-th row and j-th column of a matrix $\mathcal{T}$, as $(\mathcal{T})_{ij}$. We will use the notation $\underline{n} = \{1, 2, \dots, n\}$ with $n \in \mathbb{N}$ for sets. Cardinality of set S as $|S|$.

◯ ◊

2.2 Matheteil

Linear Order Measurable set (used in Definition 2.19) Iverson Brackets

Definition 2.1 (Gershgorin circle theorem). *Let $\mathcal{T} \in \mathbb{C}^{n\times n}$ be a square complex matrix, then for every eigenvalue λ of $\mathcal{T}$ it is true that*

$$|\lambda - (\mathcal{T})_{ii}| \leq \sum_{j=1\wedge i\neq j}^{n} |(\mathcal{T})_{ij}| \,,$$

for any $i \in \underline{n}$ and where $|\cdot|$ denotes the absolute value, i.e. magnitude.
In other words, all eigenvalues of $\mathcal{T}$ lie on a disk in the complex plane centered at $(\mathcal{T})_{ii}$ with radius $\sum_{j=1\wedge i\neq j}^{n} |(\mathcal{T})_{ij}|$.

Matrix invertible if 1 not eigenvalue

Definition 2.2 (Row-Stochastic Matrix). *A square matrix $\mathcal{T} \in \mathbb{R}_{\geq 0}^{n\times n}$ is row-stochastic, if and only if $\sum_{j=0}^{n}(\mathcal{T})_{ij} = 1$ for all rows $i \in \underline{n}$.*

When considering each row as a function $f_i(j) = (\mathcal{T})_{ij}$, these functions are discrete probability distributions.

Definition 2.3 ((Strictly) Row-Substochastic Matrix). *A square matrix $\mathcal{T} \in \mathbb{R}_{\geq 0}^{n\times n}$ is row-substochastic, if and only if $\sum_{j=0}^{n}(\mathcal{T})_{ij} \leq 1$ for all rows $i \in \underline{n}$.*
$\mathcal{T}$ is strictly row-substochastic if and only if ther exists at least one row i' with $\sum_{j=0}^{n}(\mathcal{T})_{i'j} < 1$

Theorem 2.1. *If a matrix $\mathcal{T}$ is strictly row-substochastic, then it has no eigenvalue $|\lambda| \geq 1$.*

Proof. Let us assume that $|\lambda| > 1$ is an eigenvalue of $\mathcal{T}$. As $(\mathcal{T})_{ij} \in \mathbb{R}_{\geq 0}$, using Definition 2.1

$$|\lambda - (\mathcal{T})_{ii}| = |\lambda| - (\mathcal{T})_{ii} \leq \sum_{j=1\wedge j\neq i}^{n} |(\mathcal{T})_{ij}|$$

$$\iff 1 < |\lambda| \leq (\mathcal{T})_{ii} + \sum_{j=1\wedge j\neq i}^{n} (\mathcal{T})_{ij} \,.$$

However, as $\mathcal{T}$ is strictly row-substochastic, we know that $(\mathcal{T})_{ii} + \sum_{j=1\wedge j\neq i}^{n}(\mathcal{T})_{ij} < 1 < |\lambda|$ for all $i \in \underline{n}$. Thus, any strictly row-substochastic matrix $\mathcal{T}$ has no eigenvalues $|\lambda| > 1$. □

weil $\mathcal{T}$ non negative

Definition 2.4 (Jordan normal form). *Every square matrix $\boldsymbol{A} \in \mathbb{C}^{n\times n}$ is similar to a block diagonal matrix $\boldsymbol{J} \in \mathbb{C}^{n\times n}$, also called the Jordan normal form of $\boldsymbol{A}$, meaning $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{J}\boldsymbol{P}^{-1}$ for an invertible matrix $\boldsymbol{P} \in \mathbb{C}^{n\times n}$. The blocks on the diagonal of the Jordan matrix are called Jordan blocks $\boldsymbol{J_i}$ and have the form*

$$\boldsymbol{J_i} = \begin{bmatrix} \lambda_i & 1 & 0 & \dots & 0 \\ 0 & \lambda_i & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 & 0 \\ 0 & \ddots & 0 & \lambda_i & 1 \\ 0 & 0 & \dots & 0 & \lambda_i \end{bmatrix} ,$$

where λ_i is the i-th eigenvalue of $\boldsymbol{A}$.

andere buchstaben für matrizen verwenden

Theorem 2.2 (Convergence of strictly row-substochastic matrices). *If a matrix $\boldsymbol{\mathcal{T}}$ is strictly row-substochastic, then it is convergent, i.e. $\lim_{k\to\infty} \boldsymbol{\mathcal{T}}^k = 0$*

Proof. Given a strictly row-substochastic matrix $\boldsymbol{\mathcal{T}}$, consider its Jordan normal form $\boldsymbol{\mathcal{T}} = \boldsymbol{P}\boldsymbol{J}\boldsymbol{P}^{-1}$, see Definition 2.4. Thus, $\lim_{k\to\infty} \boldsymbol{\mathcal{T}}^k = \boldsymbol{P} \lim_{k\to\infty} \boldsymbol{J}^k \boldsymbol{P}^{-1}$. Because $|\lambda| < 1$ for all eigenvalues of $\boldsymbol{\mathcal{T}}$, $\lim_{k\to\infty} \boldsymbol{J}^k = 0$. Thus, $\lim_{k\to\infty} \boldsymbol{\mathcal{T}}^k = 0$, i.e. $\boldsymbol{\mathcal{T}}$ is convergent. □

explain more in detail?

Definition 2.5 (Matrix inversion by the Neumann Series). *Given a matrix $\mathcal{T} \in \mathbb{R}^{n\times n}$ with $\lim_{k\to\infty} \mathcal{T}^k = 0$, i.e. $\mathcal{T}$ is convergent, the Neumann series $\sum_{k=0}^{\infty} \mathcal{T}^k$ converges against $(\boldsymbol{I} - \mathcal{T})^{-1}$.*

Definition 2.6 (Irreducible and Reducible Matrix). *A non-negative square matrix $\mathcal{T} \in \mathbb{R}_{\geq 0}^{n\times n} > 0$ with $n \geq 2$ is irreducible if and only if $\sum_{k=0}^{n-1} \mathcal{T}^k > 0$. If $\mathcal{T}$ is not irreducible, it is reducible.*

Definition 2.7 (Permutation Matrix). *A binary square matrix $P \in \{0,1\}^{n\times n}$ is a permutation matrix if and only if*

(i) $(P)_{ij} = 1 = (P)_{i'j}$ *with* $i \neq i'$
(ii) $(P)_{ij} = 1 = (P)_{ij'}$ *with* $j \neq j'$
$(P^{-1} = P^{\mathsf{T}})$

In other words exactly one entry of P is 1 per row and column.8

Definition 2.8 (LU-decomposition). *Let $\mathcal{T}$ be a square matrix, then (and only then) there exists an upper triangular matrix U and a lower triangular matrix L, such that*

$$\mathcal{T} = L \cdot U \ .$$

By convention, L is made unitriangular, meaning the entries on its diagnonal are equal to 1. L & U invertible: all triangular matrices inv. (if all $a_{ii} \neq 0$)

define upper/lower triang. matrices

Definition 2.9 (Conjugation of Reducible Matrices). *Let $\mathcal{T}$ be a reducible matrix, then (and only then) there exists a permutation matrix $P \in \{0,1\}^{n\times n}$ such that the matrix product $P\mathcal{T}P^{\mathsf{T}}$ is a block upper triangular matrix*

$$\mathcal{T} = \begin{bmatrix} \mathcal{T}_{11} & \mathcal{T}_{12} & \dots & \mathcal{T}_{1m} \\ 0 & \mathcal{T}_{22} & \dots & \mathcal{T}_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \mathcal{T}_{mm} \end{bmatrix}$$

wiki quelle

m submatrices

Blocks $\mathcal{T}_{ii}$ communicating groups/classes (in graph) + $_{ij}$ communicating between groups i&j??

Definition 2.10. *Invertible matrices have full rank*

Expected Values

Definition 2.11 (Discrete Probability Distribution). *A function P is a discrete probability distribution, if $P : X \to [0,1]$ where X is a countable set and $\sum_{x\in X} P(x) = 1$.*

2.3 Markov Chains

OMISSION OF $^{\mathcal{D}}$

In this Section, we provide definitions and ... for Markov Chains, as well as Interval Markov Chains. The notation is based on [1], [2]...

2.3.1 Discrete-Time (Interval) Markov Chain

Definition 2.12 (DTMC). *A discrete-time Markov chain (DTMC) is a tuple $\mathcal{D} = (S, \mathbf{P}, \iota_{init})$, where*

(i) *S is a countable, nonempty set of states,*
(ii) $\mathbf{P} : S \times S \to [0,1]$ *is the transition probability function, such that* $\sum_{s'\in S} \mathbf{P}(s,s') = 1$ *for all states* $s \in S$, *and*
(iii) $\iota_{init} : S \to [0,1]$ *is the initial distribution, such that* $\sum_{s\in S} \iota_{init}(s) = 1$.

We assume in this thesis, that the DTMCs are finite, i.e., that the set of states S is finite. Furthermore, we assume an arbitrary but fixed linear order $\prec$ on the set S, such that the states $s \in S$ can be indexed. Thus, $s_i \prec s_j$ if $s, s' \in S$ and $i < j$. Due to the conditions (i) and (ii) provided in Definition 2.12, the transition probability function $\mathbf{P}$ and the initial distribution ι_{init} are discrete probability distributions.

??(notion of a current state) Moreover, if $\mathbf{P}(s,s') > 0$ for two given $s, s' \in S$, then it is possible to move from s to s' in one step. We refer to these states s' as direct successors of s and to s as a direct predecessor of all states s'. Additionally, we define a neighbour set N, with $s' \in N(s)$, if and only if

Definition 2.13 (Transition Matrix). *The transition matrix $\mathcal{T}^{\mathcal{D}}$ of a DTMC $\mathcal{D}$ is a square matrix with size $|S| \times |S|$, such that $(\mathcal{T}^{\mathcal{D}})_{ij} = \mathbf{P}(s_i, s_j)$.*

gemeint S klar?

Notice that due to the condition, given in (ii) of Definition 2.12, that $\sum_{s_j\in S} \mathbf{P}(s_i, s_j) = 1$ for all states $s \in S$, by defintion $\sum_{j=0}^{|S|}(\mathcal{T})_{ij} = 1$ for all rows $i \in \underline{n}$. Thus, the transition matrix $\mathcal{T}$ of a DTMC $\mathcal{D}$ is row-stochastic, see Definition 2.2.

Theorem 2.3 (k-step Transition Probability). *The k-th power of the transition matrix $(\mathcal{T}^k)_{ij}$ characterizes the probability of transitioning from state i to j in precisely k steps.*

proof necessary?

Definition 2.14 (Induced weighted graph). *Every DTMC $\mathcal{D}$ induces a directed weigthed graph $\mathcal{G}^{\mathcal{D}} = (V, E, f)$, with the nodes $V = S^{\mathcal{D}}$, the edges $(s,s') \in E$ if and only if $\mathbf{P}(s,s') > 0$, and the weight function $f(s,s') = \mathbf{P}(s,s')$*

If $\mathcal{T}^{\mathcal{D}}$ is reducible, then $\mathcal{G}^{\mathcal{D}}$ is not strongly connected.

simply use **P** instead of f?

Proof. In the other direction, if $(\boldsymbol{A}^{k+1})_{ij} > 0$, then there must be at least one $l \in \underline{n}$, such that $(\boldsymbol{A}^k)_{il} \cdot (\boldsymbol{A})_{lj} > 0$. Thus, using the induction hypothesis, there must exist a path of length k from i to l and $(l,j) \in E(\mathcal{G})$. Therefore, there exists a path of length $k+1$ from i to j over l. It trivially suffices to ensure this property for $k < n$, as if a path of length $k \geq n$ exists, then at least $n+1$ nodes are visited, i.e., at least one node is visited multiple times. □

Definition 2.15 (Path of a DTMC). *An infinite path π in a DTMC $\mathcal{D}$ is a sequence $\pi = s_i \to s_j \to \dots$ of states with $s_i, s_j \in S$, such that $\mathbf{P}(\pi[k], \pi[k+1]) > 0$, where $\pi[k] \in S$ denotes the k-th element of the sequence/path for all $k \in \mathbb{N}_0$. A finite path is a finite prefix of an infinite path.*

Prob distr. of paths Furthermore, we let PathsD inf and PathsD fin represent the sets that contain infinite and finite paths π in the DTMC D, respectively. (Satz abändern, ist von Hannah)

Definition 2.16 (Probability Space of a DTMC). *The probability space of a DTMC $\mathcal{D}$ is defined as $(Paths^{\mathcal{D}}_{inf}, \sum_{Paths^{\mathcal{D}}_{inf}}, Pr^{\mathcal{D}})$, where*

- $\sum_{Paths^{\mathcal{D}}_{inf}}$ *denoted the smallest σ-algebra containing the cylinder sets $Cyl(\pi')$ for all finite paths $\pi' \in Paths^{\mathcal{D}}_{fin}$ with*

$$Cyl(\pi') = \{\pi \in Paths^{\mathcal{D}}_{inf} | \forall k \in \underline{|\pi'|} : \pi[k] = \pi'[k]\}$$

- $Pr^{\mathcal{D}}$ *is the unique probability measure on $(Paths^{\mathcal{D}}_{inf}, \sum_{Paths^{\mathcal{D}}_{inf}})$ satisfying*

$$Pr^{\mathcal{D}}(Cyl(\pi')) = \iota^{\mathcal{D}}_{init}(\pi'[0]) \cdot \prod_{k=0}^{|\pi'|-1} \mathbf{P}^{\mathcal{D}}(\pi'[k], \pi'[k+1])$$

for all finite paths $\pi' \in Paths^{\mathcal{D}}_{fin}$

Define σ-algebra, bei Hannah Def. 2.12 ff

Now that we have defined the probability space for DTMCs, we can prove Theorem 2.3.

Proof. We prove this theorem by induction. By our definition, see Definition 2.13, $(\mathcal{T})_{ij} = Pr(\pi[1] = j | \pi[0] = i) = \mathbf{P}(s_i, s_j)$, thus the theorem holds for $k = 1$. Now, assume that the theorem holds for any $k \in \underline{n}$. As $(\mathcal{T})_{ij}^{k+1} = (\mathcal{T}^k \cdot \mathcal{T})_{ij} = \sum_{m=1}^{n} (\mathcal{T}^k)_{im} \cdot (\mathcal{T})_{mj}$. Using our induction hypothesis, $(\mathcal{T}^k)_{im}$ characterizes the probability of transition from state i to m in k steps, for all $m \in \underline{n}$, i.e. $(\mathcal{T}^k)_{im} = Pr(\pi[k] = m | \pi[0] = i)$. Thus, $(\mathcal{T})_{ij}^{k+1} = \sum_{m=1}^{n} Pr(\pi[k] = m | \pi[0] = i) \cdot Pr(\pi[k+1] = j | \pi[k] = m)$ □

define conditional prob. schreibweise in mathe teil

Definition 2.17 ((Step-Bounded) Constrained Reachability (Probabilities?)). *Let $T, R \subseteq S^{\mathcal{D}}$ be two sets of states in a DTMC $\mathcal{D}$. We now define the probability of only visiting states in R until eventually a state in T is visited as $Pr(R \cup T)$, where*

$$R \cup T = \{\pi \in Paths_{inf}^{\mathcal{D}} | \exists k \in \mathbb{N} : \pi[k] \in T \wedge \forall i < k : \pi[i] \in R\}$$

Furthermore, the probability of only visint states in R until a state in T is visited at step $k \in \mathbb{N}$ is given by $Pr(R \cup^{=k}) \ldots$ (step-bounded defs necessary?)

finish proof

Definition 2.18 (Reachability Probability). *The reachability probability of a set $T \subseteq S^{\mathcal{D}}$ in a DTMC $\mathcal{D}$ is given by $Pr(\Diamond T)$, where $\Diamond T = S \cup T$*

As $S^{\mathcal{D}}$ defines all states in $\mathcal{D}$, this simply defines the probability of visiting a state $t \in T$ at an arbitrary step in the future, i.e. if $Pr(\Diamond T) > 0$ at least one state $t \in T$ is reachable. In case $T = \{t\}$ is a singleton set, we omit the braces and denote $\Diamond t$ instead of $\Diamond\{t\}$ (and $S \cup t$ instead of $S \cup T$)

how to define/denote initial state. Unterscheiden, wenn t initial oder nicht

$\sum_{Paths_{inf}^{\mathcal{D}}}$ DEFINIEREN (?)

We use the LTL unary prefix operator ◯ to denote the temporal modality "next", meaning

Definition 2.19 (Next Step Probability). *Let $\mathcal{D}$ be a DTMC and let $E \in Paths_{inf}^{\mathcal{D}}$ be a (countable) set of infinite paths in $\mathcal{D}$. We now define:*

$$\bigcirc E = \{\pi[2] \to \cdots \in Paths_{inf}^{\mathcal{D}} | \pi[1] \to \pi[2] \to \cdots \in E\}$$

The next step probability with respect to E is then defined as

$$Pr(\bigcirc E) = Pr()$$

Definition 2.20 ((Bottom) Strongly Connected Component). *A set of states $C \subseteq S^{\mathcal{D}}$ of a DTMC $\mathcal{D}$ is strongly connected if and only if for every pair of states $s, t \in S$, there exists a path $\pi \in Paths_{inf}^{\mathcal{D}}$, such that $\pi[0] = s$ and $\pi[k]$ for some $k \in \mathbb{N}$. If C is strongly connected and no proper subset, i.e. $C' \subset C$, is strongly connected, then C is called a strongly connected component (SCC). Furthermore, an SCC C is called a bottom stronlgy connected component (BSCC) if there exists no $k \in \mathbb{N}$ and no path $\pi[0] \to \cdots \to \pi[k] \in Paths_{inf}^{\mathcal{D}}$ such that $\pi[0] \in C$ and $\pi[k] \notin C$.*

Difference Paths and $\sum_{Paths}$. What is necessary here, seems to be that both were

In other words, once the model enters a BSCC, it cannot leave it again.

SCC (minimal amount of states of set such that still strongly connec) (Acyclic DTMC)

The states can be classified ...

Definition 2.21 (Transient and Recurrent states). *A state $s \in S$ is called recurrent if and only if $Pr_s(\bigcirc\Diamond s) = 1$. All states that are not recurrent are called transient.*

explain: meaning of notation Pr_s

In other words, a state is recurrent precisely when we are guaranteed to reach it from all of its direct successors, meaning once we reach a recurrent state, we are bound to return to it. (Thus reaching it infinitely often) Using these definitions, we can split the states S of a DTMC into two sets with S_{tr} containing all of the transient states and S_{re} containing all of the recurrent states.

Definition 2.22 (Absorbing state). *A states s of a DTMC $\mathcal{D}$ is refered to as an absorbing state if and only if $\mathbf{P}^{\mathcal{D}}(s, s) = 1$.*

This means, once the model enters an absorbing state, it cannot leave it. Notice, that absorbing states are a special type of recurrent states, as the only direct successor of an absorbing state $s \in S$ is the s itself which trivially has $Pr_s(\Diamond s) = 1$.

Using BSCCs we can define a special class of DTMCs.

Definition 2.23 (Absorbing DTMC). *A DTMC $\mathcal{D}$ is called absorbing if and only if all BSSCs are singleton sets consisting of an absorbing state.*

Definition 2.24 (Regular DTMC). *(?)*

Definition 2.25 (Reducible DTMC). *A DTMC $\mathcal{D}$ is called reducible if and only if its transition matrix $\mathcal{T}$ is reducible.*

As aforementioned, the induced graph of a DTMC with a reducible transition matrix is not strongly connected Reachability in graph (not strongly connected describes precisely markov chain that can be split in non empty recurrent and transient sets) $\mathcal{T}$ reducible –¿ upper triangle Definition 2.9 (–¿ proof for canonical form)

Theorem 2.4 (Limit Behaviour of DTMCs(?)). *For any DTMC $\mathcal{D}$, $s \in S$ with $\iota_{init}(s) > 0$ it holds that $Pr(\Diamond S_{re}) = 1$*

Definition 2.26 (Stopping Time). *A random variable $T : Paths_{inf}^{\mathcal{D}} \to \mathbb{N} \cup \{\infty\}$ is called a stopping time if, for every $n \geq \mathbb{N}$,*

Definition 2.27 (Strong Markov Property).

Definition 2.28 (Canonical Form). *For a DTMC $\mathcal{D}$, its transition matrix $\mathcal{T}$ is said to be in canonical form if and only if*

$$\mathcal{T} = \begin{bmatrix} Q & R \\ 0 & S \end{bmatrix}$$

where

- *(i) $Q = (\mathbf{P}(s,t))_{s,t \in S_{tr}}$ contains the transition probabilities between the transient states*
- *(ii) $R = (\mathbf{P}(s,t))_{s \in S_{tr}, t \in S_{re}}$ contains the transition probabilities between transient and recurrent state pairs*
- *(iii) $S = (\mathbf{P}(s,t))_{s,t \in S_{re}}$ contains the transition probabilities between the transient states*

For absorbing DTMCs $S = \boldsymbol{I}$. As mentioned in Definition 2.9, this is precisely possible for

(Reward Structure)

2.3.2 Expected Visiting Times for DTMCs

Intuitively, the notion of EVTs describes how often we expect the model to be in a given state. To formally introduce EVTs, we fix a DTMC $\mathcal{D}$ and start by introducing *visiting times* along paths.

Definition 2.29 (Visiting Times of Infinite Paths)**.** *The function $vt_s : Paths^{\mathcal{D}}_{inf} \to \mathbb{R}$ describes the visiting time of a state $s \in S$ along infinite paths in $\mathcal{D}$, where for $\pi \in Paths^{\mathcal{D}}_{inf}$*

$$vt_s(\pi) = \sum_{k=0}^{\infty} [\![\pi[k] = s]\!]$$

necessary for well-defined. Simply refer to her thesis?

Borel Measurability

$\to$ Thus, the Lebesgue integral of the non-negative function vt_s is well-defined for all states $s \in S^{\mathcal{D}}$. With this function, we can now define the EVT of a state s as the expected value of vt_s.

Definition 2.30 (Expected Visiting Times in DTMCs)**.** *For a state $s \in S$, the expected visiting time of s is characterized as*

$$\mathbb{E}^{\mathcal{D}}[vt_s] = \int_{\pi \in Paths^{\mathcal{D}}_{inf}} vt_s(\pi) dPr^{\mathcal{D}}(\pi) \ .$$

Additionally, we denote the expected visiting time that is obtained by fixing a certain state $t \in S$ as the only initial state as $\mathbb{E}^{\mathcal{D}}_t[vt_s]$. It is defined as

Define Expected Values in Mathe-teil (Def 2.16 bei Hannah)

$$\mathbb{E}^{\mathcal{D}}_t[vt_s] = \int_{\pi \in Paths^{\mathcal{D}}_{inf} \wedge \pi[0]=t} vt_s(\pi) dPr^{\mathcal{D}}(\pi) \ .$$

As we defined the probability measure $Pr(\pi)$ through a multiplication with the factor $\iota^{\mathcal{D}}_{init}(\pi[0])$ (see definition Definition 2.16), it is nonzero only for paths that begin in a states s with $\iota^{\mathcal{D}}_{init}(s) > 0$. In other words, the probability measure takes the initial distribution into account and thus the value $\mathbb{E}^{\mathcal{D}}[vt_s]$ gives the number of times the state s is expected to be visited from all initial states (, specified by the initial distribution).

Whenever it is clear from the context, we will omit the DTMC $\mathcal{D}$ from $\mathbb{E}^{\mathcal{D}}[vt_s]$.

Using the aforementioned classification of the states of $\mathcal{D}$ into transient and recurrent states in Definition 2.21, we can already give a characterization for the latter.

Lemma 2.1 (EVTs of Recurrent States)**.** *For all recurrent states $s \in S_{re}$ in $\mathcal{D}$, it holds that*

If needed: Define EVTs from fixed starting point

$$\mathbb{E}[vt_s] = \begin{cases} \infty & \text{if } s \text{ is reachable,} \\ 0 & \text{else.} \end{cases}$$

Thus, we will focus on finding the non-trivial expected visiting times of the transient states.

We, will now see how the expected visiting times can be calculated through a linear equation system. For this, we will first define the *fundamental matrix* which was introduced by Kemeny and Snell [3].

Definition 2.31 (Fundamental Matrix)**.** *For a DTMC $\mathcal{D}$ with the matrix $Q = (\mathbf{P}^{\mathcal{D}}(s,t))_{s,t \in S_{tr}}$ containing the transition probabilities between the transient states (defined in Definition 2.28), the fundamental matrix F is defined as,*

Use/Reference hannahs proof

$$F = (I - Q)^{-1} \ .$$

Theorem 2.5. *The matrix Q containing the transition probabilities between transient states, see Definition 2.28 is strictly row-substochastic.*

Proof. First, notice that there always needs to exist at least one recurrent state. We will prove this through a proof by contradiction. Assume that a Markov Chain $\mathcal{D}$ consists only of transient states. According to Definition 2.21 this means, that $Pr_s(\bigcirc\Diamond s) < 1$ for all states $s \in S$. Thus there exists a path $\pi \in Paths^{\mathcal{D}}_{inf}$ for all $s \in S$ such that for some $k \in \mathbb{N}$ $Pr_{\pi[k]}(\bigcirc\Diamond s) = 0$, i.e. after step k it is impossible to return to state s. However, as the state $\pi[k]$, by our assumption, is transient as well, there need to exist $\pi' \in Paths^{\mathcal{D}}_{inf}$ and $k' \in \mathbb{N}$ such that $Pr_{\pi'[k']}(\bigcirc\Diamond\pi[k]) = 0$. By repeating this argument (at most $|S|$ times), we arrive at an contradiction, where we necessarily transition to a state that is no longer reachable. Thus the statement is proven. Next, notice that there always needs to exist a $s \in S_{tr}$ and $s' \in S_{re}$ such that $\mathbf{P}(s,s') > 0$, i.e. there exists at least one transient state that as an edge to a recurrent state. By assumption all states in any given DTMC $\mathcal{D}$ are reachable and thus also a transient state $s \in S_{tr}$. Using the same argument as for the first step, we can conclude that a recurrent state $s' \in S_{re}$ must be reachable from s, i.e. $Pr_s(\bigcirc\Diamond) = 1$. Thus there necessarily exists an edge from transient to recurrent. Through condition (ii) of Definition 2.12, we know that $\sum_{j=0}^{n}(Q)_{ij} \leq 1$ for all $i \in |S_{tr}|$, i.e. Q is row-substochastic. As at least one transient state s has an edge $\mathbf{P}(s,s') > 0$, with $s' \in S_{re}$, it follows that $\sum_{j=0}^{n}(Q)_{sj} < 1$,i.e. Q is strictly row-substochastic. □

perhaps do induction argument instead, maybe cleaner?

As Q is strictly row-substochastic, we can thus conclude using Theorem 2.2, that Q is also convergent. Therefore, the Neumann series of Q converges with $\sum_{k=0}^{\infty} Q^k = (\boldsymbol{I} - Q)^{-1}$, i.e. the fundamental matrix (see Definition 2.31) is well-defined.

Furthermore, in [3] Kemeny and Snell prove that each entry $(F)_{ij} = \mathbb{E}_i[vt_j]$. From this equation, it intuitively follows that transient states are only visited a finite amount of times. Moreover, we can use this result to define a general characterization using the initial distribution.

(if $S_{tr} \neq \emptyset$, else Q = 0 and trivially strictly row-substoch perhaps mention at beginnign)

Theorem 2.6 (Calculating EVTs using the Fundamental Matrix)**.** *Given a DTMC $\mathcal{D}$ and the corresponding matrix Q, we can define*

$$\mathbb{E}[vt_s] = \sum_{j=0}^{n} (F)_{js} \cdot \iota_{init}(j)$$

Proof. As defined in Definition 2.30,

$$\mathbb{E}^{\mathcal{D}}[vt_s] = \int_{\pi \in Paths^{\mathcal{D}}_{inf}} vt_s(\pi) dPr^{\mathcal{D}}(\pi) \ .$$

We now substitute the visit time function $vt_s(\pi)$, using Definition 2.29:

perhaps quote

$$\mathbb{E}^{\mathcal{D}}[vt_s] = \int_{\pi \in Paths^{\mathcal{D}}_{inf}} \sum_{k=0}^{\infty} [\![\pi[k] = s]\!] dPr^{\mathcal{D}}(\pi)$$

$$\overset{\substack{\text{Mon. Con. Th.} \\ vt_s \text{ is non-negative}}}{\iff} \sum_{k=0}^{\infty} \int_{\pi \in Paths^{\mathcal{D}}_{inf}} [\![\pi[k] = s]\!] dPr^{\mathcal{D}}(\pi) \ .$$

how to do enumeration

If

Since $\int_{\pi \in Paths^{\mathcal{D}}_{inf}} [\![\pi[k] = s]\!] dPr^{\mathcal{D}}(\pi) = Pr^{\mathcal{D}}(\pi_k = s)$, we can thus conclude that:

$$\mathbb{E}^{\mathcal{D}}[vt_s] = \sum_{k=0}^{\infty} Pr^{\mathcal{D}}(\pi[k] = s)$$

here $Pr^{\mathcal{D}}(\pi[k] = s)$ means: prob that that some path in s at step k

Using the *Law of Total Probability*, we can characterize the probability $Pr^{\mathcal{D}}(\pi[k] = s)$ by conditioning it on the possible initial states. Thereby, $Pr^{\mathcal{D}}(\pi[k] = s) = \sum_{j=0}^{n} Pr^{\mathcal{D}}(\pi[k] = s | \pi[0] = j) \cdot Pr^{\mathcal{D}}(\pi[0])$, where $Pr^{\mathcal{D}}(\pi[0] = j) = \iota_{init}(j)$. We now substitute $Pr^{\mathcal{D}}(\pi[k] = s)$ by this term:

$$\mathbb{E}^{\mathcal{D}}[vt_s] = \sum_{k=0}^{\infty} \sum_{j=0}^{n} Pr^{\mathcal{D}}(\pi[k] = s | \pi[0] = j) \cdot \iota_{init}(j)$$

$$\underset{\text{non-neg. terms}}{\overset{\cdots}{\iff}} \sum_{j=0}^{n} \left(\sum_{k=0}^{\infty} Pr^{\mathcal{D}}(\pi[k] = s | \pi[0] = j) \right) \cdot \iota_{init}(j)$$

Using Theorem 2.3, we can now substitute $Pr^{\mathcal{D}}(\pi[k] = s | \pi[0] = j) = (Q^k)_{js}$. Thus, $\sum_{k=0}^{\infty} Pr^{\mathcal{D}}(\pi[k] = s | \pi[0] = j) = \sum_{k=0}^{\infty} (Q^k)_{js} = (F)_{js}$. Finally, we can derive the final equation

include in math-eteil

$$\mathbb{E}[vt_s] = \sum_{j=0}^{n} (F)_{js} \cdot \iota_{init}(j)$$

□

As $(F)_s = ((I - Q)^{-1})_s$, see Definition 2.31, we can rewrite Theorem 2.6, such that

add to math-eteil: non-neg. terms & cond. prob-ability no-tation

$$\mathbb{E}[vt_s] = ((I-Q)^{-1})_s \cdot \iota_{init} \iff \mathbb{E}[vt_s] = \mathbb{E}[vt_s] \cdot Q + \iota_{init} ,$$

where $\iota_{init} = (\iota_{init}(s_1), \ldots, \iota_{init}(s_n))^{\mathsf{T}}$ for all $s \in S_{tr}$.

As the calculation of the inverse of a high-dimensional matrix (for large models/chains) becomes infeasible, we will take a look at how the EVTs can be calculated without it.

For this, we define $\mathcal{E} = (\mathbb{E}[vt_{s_1}], \ldots, \mathbb{E}[vt_{s_n}])^{\mathsf{T}}$

$\mathcal{E} = (I - Q^{\mathsf{T}})^{-1} \cdot \iota_{init}$

$(I - Q^{\mathsf{T}}) \cdot \mathcal{E} = \iota_{init}$

$e - Q^{\mathsf{T}} \cdot \mathcal{E} = \iota_{init}$

$$\mathcal{E} = \iota_{init} + Q^{\mathsf{T}} \cdot \mathcal{E} \tag{1}$$

$(\mathcal{E})_s = \iota_{init}(s) + (Q^{\mathsf{T}})_s \cdot \mathcal{E} \iff \mathbb{E}[vt_s] = \iota_{init}(s) + \sum_{t \in S_{tr}} \mathbf{P}(t,s) \cdot \mathbb{E}[vt_t]$ (where $(Q^{\mathsf{T}})_s$ is the row corresponding to state s)

As this equation is iterative, we will now briefly prove that it is finite. For this, we will use LU-decomposition, see Definition 2.8. Thus, we can rewrite

Umformung länger ausführen?

$$\mathcal{E} = \iota_{init} + Q^{\mathsf{T}} \cdot \mathcal{E} = \mathcal{E} = \iota_{init} + (LU)^{\mathsf{T}} \cdot \mathcal{E}$$
$$= \iota_{init} + U^{\mathsf{T}} \cdot L^{\mathsf{T}} \cdot \mathcal{E}$$

Using the triangular property of U and L, we can now show, that the computation is finite.

Proof. Since L^{T} is an upper triangular matrix, we can define $(L^{\mathsf{T}}e)_i = \sum_{i=j}^{n} (L^{\mathsf{T}})_{ij} j \cdot e_j = e_i + \sum_{j=i+1} (L^{\mathsf{T}})_{ij} \cdot e_j$ (making nutze, dass $(L)_{ii} = 1$ use of $(L_{ii}) = 0$, as L is unitriangular).

And thus since U^{T} is a lower triangular matrix, $e_i = \iota_{init}(s_i) + \sum_{j=1}^{i} (U^{\mathsf{T}})_{ij} \cdot (L^{\mathsf{T}}e)_j = \iota_{init}(s_i) + \sum_{j=1}^{i} (U^{\mathsf{T}})_{ij} \cdot (e_j + \sum_{k=j+1}^{n} (L^{\mathsf{T}})_{kj} \cdot e_k)$. □

Assoziativtät gegeben

In the rest of this thesis, we will use the shorter vector notation $\mathbb{E}[vt_s] = \iota_{init}(s) + (Q)_s \cdot \mathcal{E}$ NOTICE THAT ONLY S_{tr} USED (as only those necessary for calc.) –¿ Others irrelevant

2.3.3 Relating Expected Visiting Times and Expected Total Rewards

2.3.4 Relating Expected Visiting Times and Quantitative Reachability

Lemma 2.2. *Given a DTMC $\mathcal{D}$ and its fundamental matrix F, we can define*

$$(F)_{ts} = Pr_t(\Diamond s) \cdot (F)_{ss}$$

for all $s \neq t \in S_{tr}$

Trivially, all paths that reach s from t at least once visit s. Thus, intuitively, the *expected visiting time* of s starting in t is simply the product of the probability of reaching s and the *expected visiting time* of s starting in s.

Lemma 2.3. *Given a DTMC $\mathcal{D}$ and its fundamental matrix F, we can define*

$$(F)_{ss} = \frac{1}{1 - Pr_s(\Diamond s)}$$

for all $s \in S_{tr}$, where $Pr_s(\Diamond s) = \sum_{t \in N(s)} \mathbf{P}(s,t) \cdot Pr_t(\Diamond s)$

Proof. As the only initial state is s, it is necessarily visited at least once. After that the probability that s will be visited at least one more time is $Pr_s(\Diamond s)$ and the probability that it will not, is accordingly $(1 - Pr_s(\Diamond s))$. Moreover, the probability of visiting s exactly k times is precisely the probability of returning $k-1$ times and not return again afterwards. Thus, $Pr_s(N_s = k) = Pr_s(\Diamond s)^{k-1}(1 - Pr_s(\Diamond s))$ for $k \in \mathbb{N}$, where N_s is a random variable representing the total number of time s is visited (Pr_s means starting in s). Intuitively,

define Pr_t

$$(F)_{ss} = \sum_{k=1}^{\infty} k \cdot Pr_s(N_s = k)$$
$$= \sum_{k=1}^{\infty} k \cdot Pr_s(\Diamond s)^{k-1}(1 - Pr_s(\Diamond s))$$
$$= (1 - Pr_s(\Diamond s)) \sum_{k=1}^{\infty} k \cdot Pr_s(\Diamond s)^{k-1}$$
$$\overset{\text{geom. series}}{=} (1 - Pr_s(\Diamond s)) \cdot \frac{1}{(1 - Pr_s(\Diamond s))^2} = \frac{1}{1 - Pr_s(\Diamond s)} .$$

more concrete proof necessary?

□

Finally, using Theorem 2.6,

$$\mathbb{E}[vt_s] = \sum_{t \in S_{tr}} (F)_{ts} \cdot \iota_{init}(t)$$
$$\overset{\text{Lemma 2.2}}{=} \sum_{t \in S_{tr}} Pr_t(\Diamond s) \cdot (F)_{ss} \cdot \iota_{init}(t)$$
$$\overset{\text{Lemma 2.3}}{=} \sum_{t \in S_{tr}} \frac{Pr_t(\Diamond s)}{1 - Pr_s(\Diamond s)} \cdot \iota_{init}(t) .$$

use strong markov property somewhere in this proof; to justify prob-ability

Moreover, using Lemma 2.2 we can trivially calculate the quantitative reachability of a given state with $Pr_t(\Diamond s) = \frac{(F)_{ts}}{(F)_{ss}}$. To calculate the quantitative reachability of a given state without a restriction of the initial state: $Pr_t(\Diamond s) = \frac{1}{(F)_{ss}} \cdot \sum_{t \in S_{tr}} (F)_{ts} \cdot \iota_{init}(t)$

Show LU constructed from Q –¿ such that connection between LU and Q defined; not necessary because literally Q=LU

2.3.5 Relating Expected Visiting Times and Hitting Times

Definition 2.32 (Hitting Times)**.**

2.3.6 Consideration for limiting relevant DTMC state size

In Section 2.3.2, we concluded that due to Lemma 2.1, only the transient states need to be considered for the calculation of the EVTs.

2.4 Interval Markov Chains

In this thesis, we will focus on Interval Markov Chains. These are defined as follows.

Definition 2.33 (Interval Markov Chain)**.** *An Interval-valued Discrete-time Markov chain (IDTMC) is a tuple* $\mathcal{I} = (S, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$*, where*

(i) S *is a countable, nonempty set of states,*
(ii) $\underline{\mathbf{P}} : S \times S \to [0,1]$ *is the transition probability function, where* $\underline{\mathbf{P}}(s,s')$ *gives the lower bound on the transition probability from state* s *to* s'*,*
(iii) $\overline{\mathbf{P}} : S \times S \to [0,1]$ *is the transition probability function, where* $\overline{\mathbf{P}}(s,s')$ *gives the upper bound on the transition probability from state* s *to* s'*,*
(iv) $\underline{\iota}_{init} : S \to [0,1]$ *is the upper bound on the initial distribution, and*
(v) $\overline{\iota}_{init} : S \to [0,1]$ *is the upper bound on the initial distribution.*

However, an IDTMC itself not usable. Interested in DTMCs that satisfy IDTMC (we denote as $\mathcal{D} \models \mathcal{I}$) There are two commonly used semantics, which we will consider in this paper. The first one is the *once-and-for-all*, see [4], [2], [5].

EVTs für IMCs

ALSO CALLED UNCERTAIN MARKOV CHAIN (UMC)/UMC SEMANTIC

Definition 2.34 (Once-and-for-all Semantic)**.** *An arbitrary DTMC* $\mathcal{D}$ *satisfies an IDTMC* $\mathcal{I}$ *if and only if* $\underline{\mathbf{P}}^{\mathcal{I}}(s,s') \leq \mathbf{P}^{\mathcal{D}}(s,s') \leq \overline{\mathbf{P}}^{\mathcal{I}}(s,s')$ *and* $\underline{\iota}^{\mathcal{I}}_{init}(s,s') \leq \iota^{\mathcal{D}}_{init}(s,s') \leq \overline{\iota}^{\mathcal{I}}_{init}(s,s')$ *for all* $s, s' \in S^{\mathcal{I}} = S^{\mathcal{D}}$*.*

Thus, any DTMC whose initial distribution and transition probabilities lie in the intervals defined by the IDTMC is satisfactory.

Definition 2.35 (IMDP semantic)**.** *An arbitrary DTMC* $\mathcal{D} = (T, \mathbf{P}, \iota_{init})$ *satisfies an IDTMC* $\mathcal{I} = (S, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$ *if and only if there exists a (surjective?) mapping* $\tau : T \to S$ *such that*

(i) $\tau(t) = s \in S$ *for all* $t \in T \implies \underline{\iota}_{init}(s) \leq \iota_{init}(t) \leq \overline{\iota}_{init}(s)$
(ii) $\tau(t) = s \in S$ *for all* $t \in T \implies \underline{\mathbf{P}}(s) \leq \mathbf{P}(t) \leq \overline{\mathbf{P}}(s)$

(Explain how they describe/define it: the minor differences between them/relation between them)

2.4.1 Expected Visiting Times for IDTMCs

As, we have discussed , in the previous subsection Section 2.4, IDTMCs can be seen as a collection of DTMCs that satisfy the interval constraints defined by the IDTMC. As only the satisfying DTMCs can be properly interpreted . . . When defining EVTs for IDTMCs it is thus useful to define them as bounds for the EVTs of the satisfying DTMCs.

discuss this somewhere in prev. subsubsection

Definition 2.36 (EVTs of IDTMCs)**.** *For a IDTMC* $\mathcal{I}$*, we define the upper expected visiting time (uEVT)* $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ *and lower expected visiting time (lEVT)* $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ *for all states* $s \in S^{\mathcal{I}}$*, as follows*

(i) $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \mathbb{E}^{\mathcal{D}}[vt_s]$*, where* $\mathcal{D} \models \mathcal{I}$ *and* $\mathbb{E}^{\mathcal{D}}[vt_s] \geq \mathbb{E}^{\mathcal{D}'}[vt_s]$ *for all* $\mathcal{D}' \models \mathcal{I}$*.*
(ii) $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \mathbb{E}^{\mathcal{D}}[vt_s]$*, where* $\mathcal{D} \models \mathcal{I}$ *and* $\mathbb{E}^{\mathcal{D}}[vt_s] \leq \mathbb{E}^{\mathcal{D}'}[vt_s]$ *for all* $\mathcal{D}' \models \mathcal{I}$*.*

2.4.2 Relating Expected Visiting Times and Expected Total Rewards

2.4.3 Relating Expected Visiting Times and Quantitative Reachability

We use the definition of the quantitative reachability property in IDTMCs provided in [6].

Given a IDTMC $\mathcal{I}$ and a set of states $\Gamma \subseteq S$, then a quantitative reachability property on $\mathcal{I}$ is defined as $Pr^{\mathcal{I}}(\Diamond\Gamma) \sim p$, where $0 \leq p \leq 1$ and $\sim \in \{\leq, <, >, \geq\}$. This quantitative property is satisfied if and only if there exists a DTMC $\mathcal{D} \models \mathcal{I}$ such that $Pr^{\mathcal{D}}(\Diamond\Gamma) \sim p$.

Given the quantitative reachabilities of the DTMCs $\mathcal{D}$ that satisfy a certain IDTMC $\mathcal{I}$, we can calculate the corresponding EVTs using Section 2.3.4. However, to determine the lEVT and uEVT of $\mathcal{I}$, we need to solve the following linear-fractional programming (LFP):

define LFP (and also NLP etc.) in Math-eteil

Definition 2.37 (LFP for uEVT of state s using quantitative reachabilities)**.**

$$\begin{aligned} \max \quad & \sum_{t \in S_{tr}} \frac{Pr^{\mathcal{D}}_t(\Diamond s)}{1 - Pr^{\mathcal{D}}_s(\Diamond s)} \cdot \iota^{\mathcal{D}}_{init}(t) & & \\ s.t. \quad & Pr_t(\Diamond s) & \mathcal{D} \models \mathcal{I} & \\ & Pr_s(\Diamond s) & & (4) \\ & \iota_{init}(s') & & (4) \end{aligned}$$

However, determining the quantitative reachabilities

with this definition: uncountable MCs –¿ thus not well-defined; need BFS or something like that

3 Calculating EVTs of IDTMCs

Most intuitive upper and lower bounds for uEVTs and lEVTs of IDTMCs can be derived by using equation (CITE) and the upper and lower transition probability functions and initial distributions defined in Definition 2.33.

Theorem 3.1 (Upper/Lower Bounds for uEVTs/lEVTs)**.** *For any IDTMC* $\mathcal{I}$*, the uEVT is bounded from above by* $\overline{\iota}_{init} + \sum_{t \in S_{tr}} \overline{\mathbf{P}}(t,s)^{\mathsf{T}} \cdot \mathbb{E}[vt_t]$ *and the lEVT is bounded from below by* $\underline{\iota}_{init} + \sum_{t \in S_{tr}} \underline{\mathbf{P}}(t,s)^{\mathsf{T}} \cdot \mathbb{E}[vt_t]$

Proof. For any IDTMC $\mathcal{I}$ there exists a DTMC $\mathcal{D}$ that satisfies $\mathcal{I}$, i.e. $\mathcal{D} \models \mathcal{I}$, and $\mathbb{E}^{\mathcal{D}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}}[vt_s]$. As $\mathcal{D} \models \mathcal{I}$ it follows that $\mathbf{P}(s,s') \leq \overline{\mathbf{P}}(s,s')$ for all $s, s' \in S$.

Thus, $\mathbb{E}^{\mathcal{D}}[vt_s] = \iota_{init} + \sum_{t \in S_{tr}} \mathbf{P}(t,s)^{\mathsf{T}} \cdot \mathbb{E}[vt_t] \leq \overline{\iota}_{init} + \sum_{t \in S_{tr}} \overline{\mathbf{P}}(t,s)^{\mathsf{T}} \cdot \mathbb{E}[vt_t] = \overline{\mathbb{E}}^{\mathcal{I}}[vt_s]$. The identical argument can be made to prove that lower bound for the lEVT. □

While these bounds can be computed easily, they are note very tight. Therefore, we will now take a look at how we can define the precise values of the uEVTs and lEVTs.

extend proof

3.1 Calculating uEVTs/lEVTs of IDTMCs using the UMC semantic

We will now have a look at how the uEVTs and lEVTs can be calculated for IDTMCs using the UMC semantic. For this, we will redefine Eq. (1).

Notice, that as there can be multiple DTMCs that satisfy an IDTMC, there can be multiple admissable transition matrices between transient states Q and initial distribution vectors ι_{init}. To find the lEVTs and uEVTs, we thus need to find the matrix Q and vector ι_{init} of a satisfying DTMC that maximize or minimze the EVT of a given state in this DTMC.

To show, that there is a

Theorem 3.2 (Calculating uEVTs with NLPs). *For any IDTMC $\mathcal{I}$, the uEVTs and lEVTs of all states $s \in S^{\mathcal{I}}$ can be calculated as follows*

$$
\begin{aligned}
\max \quad & \iota(s) + (U^{\mathsf{T}} \cdot (L^{\mathsf{T}} \cdot e))_s \\
s.t. \quad & \sum_{s_i \in N(s)} q_{s' s_i} = 1 && \forall s' \in S_{tr} && (1)\\
& \sum_{s' \in S} \iota(s') = 1 && && (2)\\
& \underline{\mathbf{P}}^{\mathcal{I}}(s', \hat{s}) \leq q_{s'\hat{s}} \leq \overline{\mathbf{P}}^{\mathcal{I}}(s', \hat{s}) && \forall s', \hat{s} \in S^{\mathcal{I}}_{tr} && (3)\\
& \underline{\iota}^{\mathcal{I}}_{init}(s') \leq \iota(s') \leq \overline{\iota}^{\mathcal{I}}_{init}(s') && \forall s' \in S^{\mathcal{I}}_{tr} && (4)
\end{aligned}
$$

Values for S_{re} chosen arbitrarily but correctly (bounds)

Theorem 3.3 (Calculating lEVTs with ILPs). *For any IDTMC $\mathcal{I}$, the uEVTs and lEVTs of all states $s \in S^{\mathcal{I}}$ can be calculated as follows*

$$
\begin{aligned}
\min \quad & \sum_{i=1}^{n} c_i x_i \\
s.t. \quad & \sum_{s_i \in N(s)} P(s, s_i) = 1 && \forall s \in S_{tr} \\
& \sum_{s \in S} \iota_{init}(s) = 1 \\
& P(s, s') \in [0, 1] && \forall s, s' \in S^{\mathcal{I}}
\end{aligned}
$$

We will now prove the optimality of these ILPs.

Proof. We will have a look at the calculation of the uEVTs through ILPs first. First, we will prove "correctness" (??) (Continue with optimality). For this, let x be a given valid (not necessarily optimal) solution of the ILP for the uEVT of a state s for a IDTMC $\mathcal{I}$ and $\mathcal{D}$ a DTMC that satisfies $\mathcal{I}$, i.e. $\mathcal{D} \models \mathcal{I}$, where $\mathbf{P}^{\mathcal{D}}(s, s') = q_{s,s'}$ and $\iota^{\mathcal{D}}_{init}(s) = \iota(s)$ for all $s, s' \in S_{tr}$.

Thus, if x is a solution, then $\sum_{\hat{s} \in N(s')} \mathbf{P}(s', \hat{s}) = 1$ for all $s' \in S_{tr}$.

Moreover, $\underline{\mathbf{P}}^{\mathcal{I}}(s', \hat{s}) \leq \mathbf{P}^{\mathcal{D}}(s, s') \leq \overline{\mathbf{P}}^{\mathcal{I}}(s', \hat{s})$ and $\underline{\iota}^{\mathcal{I}}_{init}(s') \leq \iota^{\mathcal{D}}_{init} \leq \overline{\iota}^{\mathcal{I}}_{init}(s')$ for all transitive states $s', \hat{s} \in S_{tr}$ and by definition also for all recurrent states S_{re}.

...

□

Definition 3.1 (Trace of a matrix). *For a given matrix M, its trace is defined as $tr(M) = \sum_i (M)_{ii} = \sum_i \lambda_i$, where λ_i denote the eigenvalues of M*

Definition 3.2 (Nonzero eigenvalues). *For a given matrix M, if $M \neq 0$, then M has at least one nonzero eigenvalue*

As shown before, the EVTs of a DTMC $\mathcal{D}$ can be characterized by $e = (I - Q^{\mathsf{T}})^{-1} \cdot \iota_{init}$. Let us know consider the subclass of this problem, where $Q^3 = 0$ (and $\iota_{init}(s) > 0$ for at least one $s \in S_{tr}$). For this subclass it is necessarily true, that $Q^k = 0$, for all $k \geq 3$. As $\lim_{k \to \infty} Q^k = 0$ for all Q of our problem (as Q is strictly row-substochastic and trivially especially for this subclass), we know that $(I - Q^{\mathsf{T}})^{-1} = (\sum_{k=0}^{\infty} Q^k)^{\mathsf{T}}$. Thus, for our subclass $(I - Q^{\mathsf{T}})^{-1} = (I + Q + Q^2)$ and therefore, $e = (I + Q + (Q^{\mathsf{T}})^2) \cdot \iota_{init} = (I + Q^{\mathsf{T}} + (Q^2)^{\mathsf{T}}) \cdot \iota_{init}$.

Thus, $e = \iota_{init} + \begin{bmatrix} \sum_{j=0}^{n} q_{j1} \cdot \iota_{init}(j) \\ \vdots \\ \sum_{j=0}^{n} q_{jn} \cdot \iota_{init}(j) \end{bmatrix} + \begin{bmatrix} \sum_{j=0}^{n} (\sum_{k=0}^{n} q_{jk} q_{k1}) \iota_{init}(j) \\ \vdots \\ \sum_{j=0}^{n} (\sum_{k=0}^{n} q_{jk} q_{kn}) \iota_{init}(j) \end{bmatrix}$.

Or for a single state:

$$e_t = \iota_{init}(t) + \sum_{j=0}^{n} q_{j\hat{t}} \cdot \iota_{init}(j) + \sum_{j=0}^{n} (\sum_{k=0}^{n} q_{jk} q_{k\hat{t}}) \iota_{init}(j) \ . \quad (2)$$

To determine the uEVTs and lEVTs, we are trying to find the maximum and minimum for this equation, that satisfies the constraints of the IDTMC $\mathcal{I}$.
(where $\hat{t}$ gives the index of state t in the ordering)

Definition 3.3 (NP-hard quadratic programming (QP)). *A quadratic programming problem with n variables and m constraints can be formulated as:*

find quote

$$\textit{minimize } x^{\mathsf{T}} M x + c^{\mathsf{T}} x + d$$

$$\textit{s.t. } Ax \leq b \ ,$$

where $c \in \mathbb{R}^n$, $M \in \mathbb{R}^{n \times n}$ and symmetric, $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and $d \in \mathbb{R}$.

The quadratic programming problem where M is not positive definite, i.e. M has a negative eigenvalue [7], is NP-hard

Thus, we can now rewrite Eq. (2) to fit such a quadratic programming problem. For this, we further specify our subclass to only include IDTMCs with a fixed initial distribution for the transient states, i.e. $\underline{\iota}_{init}(s) = \iota_{init}(s) = \overline{\iota}_{init}(s)$ for all $s \in S_{tr}$. We can then define $d = \iota_{init}(t)$ and $x = (q_{00}, q_{01}, \ldots, q_{0n}, q_{10}, \ldots, q_{nn})^{\mathsf{T}}$. Furthermore, we then accordingly define $M_{p,q} = \frac{1}{2} \iota_{init}(j)$, where $p = j(n+1) + k$ and $q = k(n+1) + \hat{t}$, for all $j, k \in \underline{n}$ and all other entries as zero. Lastly, we define $c_{j(n+1)+\hat{t}} = \iota_{init}(j)$ and all other entries as zero.

We will now continue by briefly showing, that this formulation is correct, i.e. $x^{\mathsf{T}} M x + c^{\mathsf{T}} x + d = \iota_{init}(t) + \sum_{j=0}^{n} q_{j\hat{t}} \cdot \iota_{init}(j) + \sum_{j=0}^{n} (\sum_{k=0}^{n} q_{jk} q_{k\hat{t}}) \iota_{init}(j)$. Firstly, trivially $d = \iota_{init}(t)$, (see its definition). Continuing, $c^{\mathsf{T}} x = \sum_{j=0}^{n} c_{j(n+1)+\hat{t}} \cdot x_{j(n+1)+\hat{t}}$, as all other entries of c are defined as zero. As $c_{j(n+1)+\hat{t}} = \iota_{init}(j)$ and $x_{j(n+1)+\hat{t}} = q_{j\hat{t}}$, thus $c^{\mathsf{T}} x = \sum_{j=0}^{n} q_{j\hat{t}} \cdot \iota_{init}(j)$. Lastly, $x^{\mathsf{T}} M x = \sum_{pq} M_{pq} x_p x_q$. When substituing the nonzero entries of M, where $p = j(n+1) + k$ and $q = k(n+1) + \hat{t}$, we get $\sum_{j,k} M_{j(n+1)+k, k(n+1)+\hat{t}} \cdot x_{j(n+1)+k} \cdot x_{k(n+1)+\hat{t}} +$

S_{re} direkt in ILP schreiben um Zeitarbeit zu sparen?? Ist halt aber nicht für Berechnung notwendig

$\sum_{j,k} M_{k(n+1)+\hat{t},j(n+1)+k} \cdot x_{k(n+1)+\hat{t}} \cdot x_{j(n+1)+k}$. As M is symmetrical and $M_{pq} = \iota_{init}(j)$ for these nonzero entries we can simplify to: $\sum_{j,k} \iota_{init}(j) \cdot x_{j(n+1)+k} \cdot x_{k(n+1)+\hat{t}}$. Now, finally substituing back $x_{j(n+1)+k} = q_{jk}$ and $x_{k(n+1)+\hat{t}} = q_{k\hat{t}}$ gets us: $x^\mathsf{T} M x = \sum_{j,k} \iota_{init}(j) q_{jk} q_{k\hat{t}} = \sum_{j=0}^{n} (\sum_{k=0}^{n} q_{jk} q_{k\hat{t}}) \iota_{init}(j)$. This objective is then, similar to Theorem 3.2, subject to the constraints $\underline{\mathbf{P}}^{\mathcal{I}}(s', s'') \leq q_{s's''} \leq \overline{\mathbf{P}}^{\mathcal{I}}(s', s'')$ for all $s', s'' \in S^{\mathcal{I}}_{tr}$. The soundness and completeness of this QP are identical to the NLP (and will thus be omitted). Thus, this subclass of the problem is NP-hard if and only if M is not positive definite. For this we will further specify our subclass to only include IDTMCs $\mathcal{I}$ with $\underline{\iota}_{init}(t) = \overline{\iota}_{init}(t) = 0$, where t is the state for which the uEVT and lEVT are calculated. With how we defined M, the entries $(M)_{ii}$ are nonzero if and only if $j = k = \hat{t}$. Therefore, (M_{ii}) is nonzero if and only if $\iota_{init}(t) > 0$ and thus for the IDTMCs of our subclass $tr(M) = 0$. Furthermore, as $M \neq 0$ (trivially as we have defined for our subclass that $\iota_{init}(s)$ for at least one $s \in S_{tr}$), we can use Definition 3.2 to reason that M must have at least one nonzero eigenvalue. Moreover, using Definition 3.1, we know that $\sum_i \lambda_i = tr(M) = 0$, necessarily M must have a negative eigenvalue. Thus, we can reason that according to [7] this subclass belongs to a class of problems that is NP-hard. This would suggest that this problem is rather hard.

3.2 Calculating EVTs of IDTMCs using the IMDP semantic

In this subsection we will look at how uEVTs and lEVTs can be calculated for the IMDP semantic.

Theorem 3.4 (Calculating uEVTs with NLPs)**.** *For any IDTMC $\mathcal{I}$, the uEVTs of all states $s \in S^{\mathcal{I}}$ can be calculated as follows*

$$
\begin{aligned}
\max \quad & \iota(s) + (U^\mathsf{T} \cdot (L^\mathsf{T} \cdot e))_s & & \\
s.t. \quad & \sum_{s_i \in N(s)} (q_{s's_i})_k = 1 & & \forall s' \in S_{tr} \ \wedge \ k \leq k_{s'} \quad (1)\\
& \sum_{s' \in S} \iota(s') = 1 & & \quad (2)\\
& (q_{s'\hat{s}})_k = 0 & & \forall k > k_{s'} \quad (3)\\
& \underline{\mathbf{P}}^{\mathcal{I}}(s', \hat{s}) \leq (q_{s'\hat{s}})_k \leq \overline{\mathbf{P}}^{\mathcal{I}}(s', \hat{s}) & & \forall s', \hat{s} \in S^{\mathcal{I}}_{tr} \ \wedge k \leq k_{s'} \quad (4)\\
& \underline{\iota}^{\mathcal{I}}_{init}(s') \leq \iota(s') \leq \overline{\iota}^{\mathcal{I}}_{init}(s') & & \forall s' \in S^{\mathcal{I}}_{tr} \quad (5)\\
& k_{s'} \in \mathbb{N} & & \forall s' \in S^{\mathcal{I}}_{tr}
\end{aligned}
$$

Values for S_{re} chosen arbitrarily but correctly (bounds) lEVTs can be calculated by minimizing the objective

Intuitively, the IMDP semantic allows for a probability distribution to be chosen at every step in each state within the constraints/bounds set by the IDTMC. In contrast to the UMC semantic, where each state in the satisfying DTMCs have a corresponding state in the IDTMC and thus a fixed probability distribution, while the IMDP semantic allows for multiple states in the satisfying DTMCs with different fixed probability distributions to be mapped to the same corresponding state in the IDTMC. We can thus imagine an IDTMC under the IMDP semantic as a Markov Decision Process.

Definition 3.4 (Markov Descision Process (MDP))**.** *[1] A Markov Decision Process (MDP) is a tuple $D = (S, Act, A, \mathbf{P}, \iota_{init})$, where*

(i) S is a countable set of states
(ii) Act is a set of actions
(iii) $A(s) \subseteq Act$ denotes the applicable actions for all states $s \in S$
(iv) $\mathbf{P} : S \times Act \times S \to [0, 1]$ is the transition probability function such that $\sum_{s' \in S} \mathbf{P}(s, \alpha, s') = 1$ for all $s \in S$ and $\alpha \in A(s)$
(v) $\iota_{init} : S \to [0, 1]$ is the initial distribution such that $\sum_{s \in S} \iota_{init}(s) = 1$

The most intuitive reduction from an IDTMC $\mathcal{I}$ under the IMDP semantic to a MDP $\mathcal{M}$ is to define $S^{\mathcal{M}} = S^{\mathcal{I}}$ and an uncoutable set $\cup_{s \in S} A(s) = Act$ of actions where each action $\alpha \in A(s)$ corresponds to a probability distribution $\underline{\mathbf{P}}^{\mathcal{I}}(s, s') \leq \mathbf{P}^{\mathcal{M}}(s, \alpha, s') \leq \overline{\mathbf{P}}^{\mathcal{I}}(s, s')$ (for $s' \in N(s)$). In other words, the actions $\alpha \in A(s)$ in a given state $s \in S$ correspond to the probability distributions that can be chosen in s in any given step. However, as even the subsets $A(s)$ can be uncoutable, this reduction is trivially not "usable", as no algorithm, such as value or policy iteration, would terminate.

Instead, it is thus argued in [8] and [2], that the local constraints of each state of these MDPs and IDTMCs, see in Definition 3.4 $(iv) - (v)$ and Definition 2.33 $(ii) - (v)$, can be used to limit the amount of probability distributions that need to be mapped(, as we only need to consider the most extreme "behaviour"). In the theory of linear programming, this is called a basic feasible solution (BFS) . In the following, we will extend the reduction from IDTMCs to MDPs (using the IMDP semantic) presented in [8] and [2]. We will refer to the constraints of the form $\sum_{s' \in N(s)} \mathbf{P}(s, s')$, where $\underline{\mathbf{P}}(s, s') \leq \mathbf{P}(s, s') \leq \overline{\mathbf{P}}(s, s')$ as $PC(s)$ (for all $s \in S_{tr}$) and those of the form $\sum_{s \in S} \iota_{init} = 1$, where $\underline{\iota}_{init}(s) \leq \iota_{init}(s) \leq \overline{\iota}_{init}(s)$ as IC (notice not dependent on s, as only one set of inequations/constraints).

Theorem 3.5 (Basic feasible solution (BFS) for $PC(s)$)**.** *A map $\theta^s : S \to [0, 1]$ is a basic feasible solution for the constraints $PC(s)$ if and only if $\{p_{ss'} = \theta^s(s') | s' \in S\}$ is a solution for $PC(s)$ and there exists a set $S' \subseteq S$ such that $|S'| \geq |S| - 1$ and for all $s' \in S'$ either $\theta^s(s') = \underline{\mathbf{P}}(s, s')$ or $\theta^s(s') = \overline{\mathbf{P}}(s, s')$.*

Theorem 3.6 (BFS for IC)**.** *A map $\phi : S \to [0, 1]$ is a basic feasible solution for the constraints IC if and only if $\{\iota_{init}(s) = \phi(s) | s' \in S\}$ is a solution for IC and there exists a set $S' \subseteq S$ such that $|S'| \geq |S| - 1$ and for all $s \in S'$ either $\phi(s) = \underline{\iota}_{init}(s)$ or $\phi(s) = \overline{\iota}_{init}(s)$.*

In other words, at least $|S| - 1$ states are set to their maximum or minimum (at most one variable chosen fittingly such that sum-constraint is fulfilled)

Geometrically, each BFS corresponds to a vertex of the polyhedron of possible solutions (given by the constraints). (In other words, we need only the corners of the polytope, created by the constraints)

BEISPIELE: EVTS FÜR IMDP KOMISCH –¿ sobald 0er Kanten –¿ EVT 0 (bzw. 1 für einer Kanten)

FINITE SET OF STATES –¿ wahrscheinlich für reduction wichtig (weil müssen BFS für alle transient states finden –¿ wenn S_{tr} infinite, dann Konstruktion nicht möglich)

Reduction: [8] and [2] . . . examples & intuition

3.3 Relation between the UMC and IMDP semantics

3.4 EVTs in Acyclic IDTMCs

Definition 3.5 (Acyclic IDTMC)**.** *An IDTMC $\mathcal{I}$ is acyclic if and only if all satisfying DTMCs $\mathcal{D}$ ($\mathcal{D} \models \mathcal{I}$) are acyclic.*

– l/uEVT is simply maximum/minimum reachability (which can be calc in P, cite paper).

3.5 EVTs in IDTMCs

3.6 Exkurs: EVTs in aIMCs

Definition 3.6 (Augmented Interval Markov Chains)**.** *An augmented Interval Markov Chain (aIMC) $\mathcal{A}$ is a tuple $(S, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init}, C)$, where $(S, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$ is an IDTMC $\mathcal{I}$ and $C \subseteq S^4$ is a set of edge equality constraints. Similar to the once-and-for-all semantic for IDTMCs, see Definition 2.34, we define, that an arbitrary DTMC $\mathcal{D}$ satisfies an aIMC $\mathcal{A}$ if it satisfies the corresponding IDTMC $\mathcal{I} = (S, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$ and $\mathbf{P}^{\mathcal{D}}(s_1, s_2) = \mathbf{P}^{\mathcal{D}}(s_3, s_4)$ for all $(s_1, s_2, s_3, s_4) \in C$. [9]*

nur grob (manches) übernommen

In other words, aIMCs extend the definition of IMCs with an "edge-constraint" which enforces certain transition probabilities to be equal. A restriction of the quantitative reachability problem, which we discussed in Section 2.3.4 and Section 2.4.3, is the qualitative reachability problem, where is restricted to $p \in \{0, 1\}$ (see Section 2.4.3).

In [9], the authors prove that the qualtitative reachability problem for aIMCs is NP-complete. Thus, as trivially the qualitative problem can be reduced to the quantitative problem, we can thus deduce that the quantitative problem for aIMCs is NP-hard. As we have shown in . . .

möglich? weil (vllt) nicht poly-nomiell um-rech-nung möglich

4 Conclusion and Outlook

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